



3DP TYCOON

Build your 3D printing empire from a single bedroom printer

Game Design Document — a browser-based idle/incremental business simulator where a single free printer grows into a full manufacturing empire.

Version documented: 1.12.1

Studio: TheBooleanJulian

Platform: Web (HTML/CSS/JS, no build step)

Live: 3dp-tycoon.thebooleanjulian.dev

Table of Contents

1.	Overview & Design Pillars
2.	Core Gameplay Loop
3.	Progression Phases
4.	Printers
5.	Products & Materials
6.	Economy: Filament, Sales & Liquidation
7.	Tech Tree & Automation
8.	Workforce & Workshop Space
9.	Marketing & Live Events
10.	Bulk Orders, Exhibitions & Commissions
11.	Achievements
12.	Prestige & Prestigium
13.	UI/UX & Onboarding
14.	Technical Architecture
15.	Save System & Cloud Save
16.	Roadmap

1. Overview & Design Pillars

3DP Tycoon is a browser-based idle/incremental simulator: the player starts with one free 3D printer in a cramped bedroom and grows the operation into a manufacturing empire, primarily through offline-friendly automation rather than twitch input.

Elevator Pitch

You inherit a single 3D printer and \$50. Print novelty items, sell them, reinvest in better printers and materials, and gradually automate every part of the business — printing, selling, finishing, marketing — until the shop runs itself and scales toward aerospace-grade manufacturing contracts.

Design Pillars

Pillar	What it means in practice
Meaningful choice, not just bigger numbers	Every printer/material/colour combination is priced independently, not blended into a fleet average — switching setups is a real decision with a visible payoff.
Always a path forward	LIQUIDATE (instant, discounted sale) exists from minute one, before any research — there's never a point where the player is stuck with product and no way to convert it to cash.
Automation is the mid-game, not the end-game	Cheap, early automation (auto-print, auto-liquidate, filament resupply) is reachable well before the late game, so the loop shifts from manual clicking to management early.
No dead currency	Every currency in the game (cash, Prestigium) has an active spend sink; nothing accumulates without a use.
Respect the player's time and trust	No punishing silent failures — bulk orders became standing goals instead of auto-failing, saves warn before overwriting, and errors degrade gracefully (e.g. cloud save with no DB configured) instead of breaking the game.

2. Core Gameplay Loop

The loop operates on three nested timescales:

Loop	Cadence	Actions
Micro loop	Seconds	Load a printer with product/material/colour, print, watch progress + ETA, resolve success/failure.
Meso loop	Minutes	Sell/liquidate stock, buy filament, react to bulk orders / exhibitions / commissions / random events, research the next affordable upgrade.
Macro loop	Session-to-session	Expand workshop space, automate more of the pipeline, hire/robotize staff, grow followers, eventually Scale Up (prestige) for a permanent multiplier and Prestigium.

The 12-step onboarding arc

1. Print with the starter Bender 3 Clone
2. LIQUIDATE first prints for cash (works before any research)
3. Buy filament in the material + colour the printer is loaded with
4. Research speed/efficiency/quality/business upgrades
5. Automate (auto-print, auto-liquidate, filament resupply — all cheap, tier 1)
6. Expand the printer fleet, unlock premium products
7. Grow workshop space (shelving → rented offices/warehouses/factories)
8. Unlock the Finishing Studio for full product value
9. Market via timelapses/exhibitions for followers & value boosts
10. Hire staff or buy robots to automate Finishing/Marketing
11. Take custom commissions, upselling 3D modelling work
12. Scale Up for a permanent multiplier + Prestigium

This arc is also taught in-game via a 6-step first-run tutorial modal and reinforced by an always-available in-game Guide (see Section 13).

3. Progression Phases

Phase is derived purely from lifetime earnings and drives flavour text/badge colour, not hard gates:

Phase	Lifetime earned	Feel
HOBBYIST	\$0 – \$1,000	Learning the ropes
MAKER	\$1,000 – \$10,000	Building momentum, first automation
BUSINESS	\$10,000 – \$100,000	A real operation; scaling begins
FARM	\$100,000 – \$1,000,000	Print farm dominance
EMPIRE	\$1,000,000+	Manufacturing empire

4. Printers

Ten printer models, each an explicit trade-off between **speed**, **quality** (a value multiplier), **cost**, filament efficiency, and fail-rate modifier — there is deliberately no single best machine. Whatever printer, material, and colour actually produced a batch determines its sale price; output is never blended into one fleet-wide average, so mixing machines and materials is a real strategic lever, not flavour text.

Printer	Cost	Speed	Quality	Notes
Bender 3 Clone	\$189 (1st free)	1.0×	Baseline	Starter; extra units cost real money to prevent a \$0-printer exploit
PrintPal Mini (Resin)	\$150	1.3×	Low	Cheap resin entry point, wasteful & finicky
Mambo Lab A1	\$800	2.5×	Above baseline	Fast CoreXY all-rounder
Proton Mono M5S (Resin)	\$2,200	3.5×	High	+60% miniature value bonus
Mambo X1 Carbon	\$3,500	5.0×	High	Multi-colour, industry standard
Boron 2.4 350	\$9,500	9.0×	Below baseline	DIY beast, temperamental
Gantry X9 Farm Unit	\$18,000	14.0×	Below baseline	Volume over finish
Helix CNC Precision Mill	\$30,000	2.0×	Very high	Slow subtractive precision
Industrial FDM	\$55,000	18.0×	High	Commercial-grade bulk production
Titan SLS Metal Printer	\$120,000	4.0×	Highest	Best-in-class metal sintering

Multi-colour Nozzles

The Nozzle branch of the tech tree (Dual/Quad-Colour Nozzle Head, Automatic Colour-Swap AI) lets a printer add up to 3 accent colours to a print for a stacking +10%/colour value bonus. Each accent colour draws a smaller top-up (15% of the primary filament cost, reducible via the Colour-Swap AI) from its own colour's stock — multi-colour output is tracked as a distinctly-tagged inventory stack rather than merged with single-colour product of the same recipe.

5. Products & Materials

Products: $20 \times 5 = 100$

Twenty base categories (Keychains through Aerospace Components) each spawn four unlockable variant tiers, so the catalogue totals 100 distinct products with no two sharing the same value/filament/print-time/unlock-cost combination.

Variant	Value	Filament	Print time	Unlock cost	Flavour
Mini	0.55×	0.5×	0.5×	0.5×	Cheap, fast early unlock
Glow-in-the-Dark	1.35×	1.15×	1.15×	1.4× + \$150	Novelty upsell
Deluxe	1.9×	1.6×	1.5×	2× + \$400	Premium finish
Custom Engraved	1.6×	1.1×	1.3×	2.5× + \$600	Biggest margins

Materials

Eight materials (six filaments, two resins) each carry a filament cost multiplier, a quality/value multiplier, and a fail-rate modifier — reflecting real-world 3D printing trade-offs (e.g. Nylon is strong but absorbs moisture and fails more; PETG is tougher and more reliable). **Carbon Fiber Composite is deliberately restricted to black only**, matching how the real material is sold — enforced both on the printer card and the filament shop.

Colour & Discrete Filament Stock

Eleven cosmetic colours each carry a small value bonus. Critically, filament stock is tracked **per material AND per colour** — a printer loaded with red PETG draws down red PETG specifically, not "PETG" in general. This is visualised as a simplified coloured "spool" graphic per material that visibly drains as stock depletes.

6. Economy: Filament, Sales & Liquidation

Design intent: there must always be a way to convert finished stock into cash, even before any research — this is why LIQUIDATE exists from the very first print.

Mechanic	Description
LIQUIDATE	Instantly sells a stack to a wholesaler at 55% of market value. Always available; the day-one income source.
Passive/demand-driven sales	Unlocked by the CraftBazaar Shop upgrade — customers buy stock automatically over time, scaled by storefront reach and publicity (followers/viral boosts).
Raw vs. finished sale value	Raw (unprocessed) prints sell for 65% of full value; running them through the Finishing Studio (sand → smooth → paint) unlocks full value.
Filament pricing	Base \$0.022/g (PLA), scaled per-material by a cost multiplier — deliberately anchored close to real-world filament pricing (~\$22/kg) for believability.

7. Tech Tree & Automation

Tech Tree

Upgrades are rendered as a node-based tech tree that branches rightward by prerequisite depth, grouped into six categories: Speed, Efficiency, Space (workshop capacity), Business, Quality, and Nozzle (multi-colour). A generic node-lane renderer computes each node's depth from its prerequisite chain and lays it out with SVG bezier connectors — the same renderer is reused for the separate Prestigium tree (Section 12).

Automation

Automation items are tiered (Basics → Advanced → Enterprise) and deliberately reachable early and cheaply, so the shift from manual to hands-off management happens well before the late game. Examples: Auto Print Queue, Filament Auto-Resupply (with a player-adjustable threshold/batch size, since late-game products can burn through the old fixed 600g default in a single print), Auto-Liquidate Overflow, Dynamic Pricing AI, Smart Restock AI, Multi-Colour Optimizer, Predictive Maintenance AI, and Demand Forecasting AI.

8. Workforce & Workshop Space

Workforce

Role	Effect when staffed
Finishing Technician / Robot Arm	Auto-feeds idle Finishing Studio stations
Social Media Manager / Autoposter Bot	Posts timelapse clips automatically
Procurement Manager / Procurement AI	–10% filament cost
Floor Supervisor / QC Sentry Drone	–12% print failure rate

Each role can be a paid worker (wage/sec, pauses automatically if the player runs out of cash) or a one-time robot purchase (no ongoing wage, but a real up-front cost and — deliberately — no refund on decommission, so it's a genuine investment decision).

Workshop Space

The player starts with room for exactly 1 printer. A cheap, early shelving upgrade line grows capacity to 8; beyond that, rented office/warehouse/mega-factory space (with ongoing rent, deducted like wages) scales capacity into the hundreds.

9. Marketing & Live Events

Marketing

Completed prints have a chance to capture a timelapse clip; posting one grants followers and a temporary value boost, with a chance to go viral for a much larger payout. Followers also grant a small permanent value bonus (capped, diminishing) and — once Social Media Presence is researched — grow slowly on their own over time, on top of active posting.

Random Events

Eleven random events swing the business beyond the scheduled bulk-order/exhibition/commission cadence: filament price swings (discount/hike), fail-rate spikes/dips, follower boosts (press features, viral screenshots), demand swings (rave reviews, shipping delays, trending hashtags), and print-speed buffs/debuffs (firmware tweaks, maintenance days). Every active timed effect is surfaced live as a chip with a countdown under the HUD, rather than only appearing as a one-off toast.

10. Bulk Orders, Exhibitions & Commissions

Bulk Orders — Standing Goals

Design note: accepting a bulk order does *not* require the stock to already exist. It becomes a tracked "Active Order" with a deadline, shown on the Products tab, auto-fulfilling the instant enough stock accumulates (or expiring if the player runs out the clock). This replaced an earlier version where acceptance silently auto-failed without existing inventory — a real usability trap.

Exhibitions

Eight exhibition types (Hobbyist Meetup up to National Manufacturing Summit) offer a paid booth in exchange for followers and a temporary value boost. Follower payoff intentionally has no hard cap — it scales with the shop's overall progress (lifetime earnings), so exhibitions stay relevant deep into the late game instead of hitting a ceiling.

Custom Commissions

Unlocked via the 3D Modelling Service upgrade — a customer requests a specific product/material/colour combination. The player can accept print-only, or upsell 3D modelling work for an extra fee, representing the CAD design behind a one-off custom part.

11. Achievements

28 achievements span the expected breadth-of-progress markers (prints completed, lifetime earnings, printers owned) and a deliberate second tier of **hard-to-attempt** achievements meant to represent genuine mastery rather than time-on-task:

- **Nothing Left To Research** — every upgrade in the tech tree
- **Fully Automated Floor** — every staff role replaced with a robot
- **One Of Every Kind** — the maximum allowed of every printer model simultaneously
- **Flawless Streak** — 100 successful prints in a row without a single failure
- **Upsell Master** — 25 commissions fulfilled with the modelling upsell
- **Master Recycler** — 10kg of filament recovered from failed prints
- **Multi-Colour Pioneer, Nozzle Master, Viral Sensation**, and more

12. Prestige & Prestigium

Scale Up (Prestige)

Once lifetime earnings cross a requirement (scaling $\times 5$ per prestige, adjustable via Prestigium research — see below), the player can Scale Up: printers, upgrades, and stock reset, but a permanent earnings multiplier (+50% per Scale Up) persists. The prestige card shows a live progress bar toward the next Scale Up rather than just raw numbers.

Prestigium — the meta-currency

Prestigium is the one thing in the game that survives every reset, by design — it's the vehicle for long-term, cross-run progression.

Every Scale Up also pays out  Prestigium — more for running further past the requirement before cashing in, rewarding patience over reflexive prestiging. Prestigium is spent on its own 10-node permanent tech tree (reusing the same node-lane renderer as the main tech tree): starting cash/filament bonuses, permanent value/speed multipliers, a lower future Scale Up requirement, bigger future Prestigium payouts, a free starting storefront, and a capstone node combining several bonuses. Every Prestigium node, once researched, has its effect re-applied to the freshly reset state on every subsequent Scale Up.

13. UI/UX & Onboarding

Element	Purpose
Tabs (Printers / Products / Upgrades / Automation / Studio / Stats)	Primary navigation, keyboard-shortcut accessible (1–6)
Always-visible printer sidebar	Every printer's job and live progress/ETA, regardless of active tab; collapses to a horizontal strip on mobile
Bufs/debufs bar	Every active timed multiplier shown with icon, label, and countdown under the HUD
Floating value text	+\$/–\$ fade-in-fade-out feedback for every sale and purchase, manual or automated
First-run tutorial	6-step modal shown once, only on a brand-new game, never on Continue
In-game Guide	Full condensed rundown of every system, reachable any time from the HUD — for players who never see the README
Settings menu	Theme (light/dark), UI scale, save/export/import, cloud save, in-game changelog, GitHub link/credit

14. Technical Architecture

Layer	Choice	Why
Frontend	Plain HTML + CSS + JavaScript, classic <code><script></code> / <code><link></code> tags	No build step — runs by double-clicking <code>index.html</code> directly, even offline
Rendering	Custom DOM-morphing renderer (<code>morph()</code> / <code>morphChildren()</code> / <code>morphNode()</code>)	Periodic re-renders patch only changed nodes, preserving focus, open dropdowns, and in-flight clicks instead of tearing down the DOM every frame
Server	Node.js — a minimal static file server plus a small <code>/api/save/*</code> JSON API	Zero framework dependency; the API is additive and never required for the core game to run
Cloud save DB	PostgreSQL (optional)	Only active if a connection is configured — degrades to local-only saves everywhere else

15. Save System & Cloud Save

- Auto-saves every 60 seconds and on tab close/hide; manual save via `Ctrl+S`
- Saves live in `localStorage` first, always — no server dependency for the core save loop
- Export/Import: portable base64 save codes via the clipboard
- **Cloud Save (optional):** a short, high-entropy code (10 characters, ~50 bits) links a save to a `cloud_saves` Postgres row (`code` , `data` JSONB , `updated_at`). Every autosave on a linked browser also pushes to the cloud. The API is rate-limited per IP and capped at 1MB per save.
- **Security note** a cloud code is a bearer secret — anyone holding it can read or overwrite that save. There is deliberately no account/login layer, trading recovery for zero-friction setup.

16. Roadmap

Idea	Notes
Sound & music	Printer hum, sale chimes, ambient workshop track with a mute toggle
Deeper mobile layout	Touch-sized controls beyond the current responsive sidebar fix
Leaderboards	Compare prestige count / net worth — needs a backend beyond cloud save
Statistics dashboard	Per-product profit graphs, print-time breakdowns
Alternate themes/skins	Beyond the existing light/dark toggle
Mod/community recipes	JSON-defined custom products/printers for self-hosters
Automated test coverage	No automated tests currently exist for the game logic itself

3DP Tycoon — dual-licensed under AGPLv3 and a commercial license (see [LICENSE / COMMERCIAL-LICENSE.md](#) in the repository). Project name/branding is not covered by either license (see [TRADEMARKS.md](#)). Document generated from the live v1.12.1 codebase and [README](#).